

Formation of Business Solutions of Cognitive Economy

I. A. Brusakova

Saint Petersburg Electrotechnical University "LETI"
St. Petersburg, Russian Federation

Abstract—Procedure of formation of adaptive business process for management of efficiency of activity of the hi-tech enterprise as procedure of management of key indicators of efficiency of the business solution with use of a service-oriented technological platform of ERP system is presented in article.

Keywords—adaptive business processes, SOA ERP, BPM-systems, KPI, metrics of effectivities, change management, knowledge management

The purpose of processing of economic measurements in a corporate information system (ERP) is formation of business solutions for an assessment of current state of efficiency of activity and for development of recommendations for further development of corporation. In the conditions of cognitive economy it is required to react to changes in activity of corporation quickly. At modernization of the knowledge-intensive productions often speed of reaction to deterioration of key indicators of efficiency will allow to avoid reengineering. For development of tools of continuous management of processes IT services, business applications, Web services which finally provide management of efficiency of activity of corporation are widely used now. Existence of the service-oriented technological platforms (SOA) of ERP allows to simplify procedures of management of changes significantly. Management of changes means management of reaction of business system to change of external and internal conditions. In case of management of technological innovations of the knowledge-intensive productions change means reorganization of technological processes of all levels of management. The acquired corporate knowledge as a non-material asset, also have to "breathe", be reconstructed depending on the made changes.

At audit of business processes methods of structuring economic information are used: notations of UML, DFD, SADT, EPC, etc. [1–3]. Representation reception the infologicheskikh of semantic models of business processes is useful [2]. In this case the subject domain of business system from formalizations of NR of business processes is presented in the form of statements which elements are business processes, their indicators, values of indicators, interrelations between business processes, between indicators, between values of indicators in a look

$$a_i = \langle \{v_m^i\}_{m=1}^M, \{\lambda_i^{i,j}\}, r^v, r^V, r^a \rangle,$$

where r^v, r^V, r^a – interrelations between indicators, values of indicators and various business processes [2].

Under corporate knowledge (CK), we will understand necessary knowledges for the description of a concrete set of realization of the main, auxiliary, through, internal business processes of corporation on a set of realization of external business processes of corporation and a set of models of conditions of carrying out monitoring of indicators of efficiency [1].

Economic measurements – values of indicators of business processes for a certain period [2].

We will enter the following designations: $\{\lambda_i^*\}_{i=1}^N$ – system of metrics of indicators of business process, where N – total of indicators; $\Delta\lambda_j^*$ – changes of values j-wow business process indicator; indicators of efficiency of business θ (for example, profit, profitability of assets, profitability in the prices specified to the real timepoint, etc.); $\bigcap\{\theta_r[\lambda_j^* \pm \Delta\lambda_j^*]\}_{r=1}^{NR}, j=1, N$ – an integrated indicator of efficiency on r-ohm to business process for a certain period, NR-total of the considered business processes.

$$\begin{aligned} I &\xrightarrow{O_1} \lambda_j^* \xrightarrow{O_3} \Delta\lambda_j^* \xrightarrow{O_4} \{\Delta\lambda_j^*\} \\ &\xrightarrow{O_5} \{\theta_r[\lambda_j^* \pm \Delta\lambda_j^*]\}_{r=1}^{NR}, j=1, N \xrightarrow{O_6} \\ &\bigcap\{\theta_r[\lambda_j^* \pm \Delta\lambda_j^*]\}_{r=1}^{NR}, j=1, N \rightarrow \\ &\xrightarrow{O_7} \overline{CK}_{ness}, \end{aligned} \quad (1)$$

where procedure of a choice of necessary structure of corporate knowledge $\overline{CK}_{ness}$, for providing the demanded indicator of efficiency of business on a set of adjustable business processes of NR for the chosen system of metrics of indicators of business processes is entered.

Adaptive business process is understood as business process for which possible borders of adjustment of indicators are estimated. The metric space of indicators of business process mnogomerno, is defined by quantity of the indicators touched as a result of management according to (2). In case

values of indicators have quantitative character, the metric space can be considered as Euclidean. If values of indicators are excellent from quantitative, it is necessary to pass to the solution of a problem of their control with use of non-Euclidean measures, for example, by Canberra, Manhattan, etc. [3, 4].

Management of knowledge, formation of business solutions – management of efficiency of activity of corporation – is realized step by step [1]:

- by results of economic measurements statements (corporate knowledge CK) in the form of the linked hierarchically ordered nominal sign received, for example, with use of ontologic approach and interpreting components of corporate knowledge in a look are formed

$$\begin{aligned} CK &= CK_{int-bp} \wedge CK_{ext-bp} \wedge \\ &CK_{coord} \wedge CK_{conditions} \wedge \\ &CK_{bus-functions} \wedge \\ &\wedge CK_{estim-eff} \end{aligned} \quad (2)$$

where – CK_{int-bp} the corporate knowledge of internal business processes gained on the basis of economic measurements of key indicators of efficiency for internal business processes; CK_{ext-bp} – corporate knowledge of external business processes; CK_{coord} – corporate knowledge of features of the organization of through business processes (changes of sequence of performance of functions, emergence new and elimination of old functions, reorganization of separate divisions);

- are formed $CK_{bus-functions}$ – corporate knowledge of features of business functions (possible changes in document flow of corporation – the standard and earlier not used documents of the company, change the property man of structure of documents, change of a place of storage of documents; possible changes in responsible for performance of functions, changes of owners of business processes, changes of the course of performance of functions);
- are formed $CK_{conditions}$ – corporate knowledge of change of models of conditions of realization of activity of corporation (a choice of other SOA architecture, other services, other bank, other virtual kernel, change of a configuration of communication networks, reorientation to other customer and the consumer, other made product, other production technology, etc.);
- are formed $CK_{estim-eff}$ – corporate knowledge about traditionally used and to attraction of new metrics of management of efficiency of business (system of the balanced indicators, KPI, BSC, metrics for management of IT services, etc.);
- corporate knowledge in the form of the statement (2) according to (1) as vectors in set-theoretic information

space of signs which coordinates are defined by concrete values of indicators of business processes that will allow to apply OLAP-technology, technology of multidimensional cubes to intellectual processing of economic information is formed;

- are formed by the linked hierarchically ordered values of indicators of business processes of the necessary structure of corporate knowledge used for ensuring efficiency of activity of corporation;
- formation for these hierarchically ordered signs of set-theoretic multidimensional information space linked nominal which dimension is defined, proceeding from dimension of the generalized relational form formed thus;
- the description of an integrated indicator of efficiency of activity of corporation θ for various structures of corporate knowledge as multidimensional interrelation of model of subject domain of the analysis of activity of corporation;
- definition for set-theoretic metric space of measures of ranging of the linked hierarchically ordered nominal signs for the solution of an optimizing problem of a choice of necessary structure of corporate knowledge for providing requirements for management of changes of activity of corporation.

We will imagine the business solution φ_t , for example, for results of IT controlling, as some statement of a look [1]:

$$\varphi_t = A_t \wedge B_t \wedge C_t \wedge D_t \wedge \dots \xrightarrow{SEM} T, \quad (3)$$

where A_t – corporate knowledge of budgeting process; B_t – corporate knowledge of results of management accounting; C_t – corporate knowledge of indicators of financial activity of corporation; D_t – corporate knowledge of intangible assets of corporation, etc.; SEM – semantic procedure of interpretation of results of the business solution which is compared, for example, with use of a linguistic scale with an element a term set of T ("well", "Unsatisfactorily", "Good").

The most used semantic procedures of interpretation of results of the business solution are procedures of formation of fuzzy productions, uses of linguistic scales [2].

Existence of a SOA technological platform allows to develop the IT services realizing the business solutions (3) and capable quickly to react to changes of parameters of business processes. The paradigm of service-oriented approach to management of information consists in possibility of "stitching" of various fragments of services, components of services, streams (compositions) of services on the basis of object-oriented approach to the analysis and design of PUSSYCATS and technologies of web services [4]. On the basis of SOA architecture the IBM WebSPHERE and SAP NetWeaver platforms are considered as known and successful platforms now.

Thus, procedure of formation of adaptive business process consists in identification of necessary structure of corporate knowledge for management of efficiency of activity of the

enterprise with use of the ERP IT services on the basis of service-oriented technological platforms.

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